

Instructions: There are a total of 5 questions. All questions are compulsory. Missing data, if any, may be suitably assumed.

1. Draw an Entity-Relationship (ER) diagram that shows the entity types, attributes, cardinality ratio, participation constraints, relationships, and specializations for the following art museum database. State any assumptions you make about the additional requirements. Assume that the following requirements were collected. [9 M]

- The museum has a collection of ART_OBJECTS. Each ART_OBJECT has a unique IdNo, an Artist, a Year, a Title, and a Description. The art objects are categorized in several ways, as discussed below.
- ART_OBJECT are categorized based on their type. There are three main types: PAINTING, and SCULPTURE, plus another type called OTHER to accommodate objects that do not fall into one of the three main types.
- A PAINTING has Paint Type (oil, watercolor, etc.), material on which it is Drawn On (paper, canvas, wood, etc.), and Style (modern, abstract, etc.).
- A SCULPTURE has Material from which it was created (wood, stone, etc.), Height, Weight, and Style.
- An art object in the OTHER category has a Type (print, photo, etc.), and Style.
- ART_OBJECT are also categorized as PERMENTENT_COLLECTION, which are owned by the museum (these have information on the Date Acquired, whether it is On Display or stored, and Cost) or BORROWED, which has information on the Collection (from which it was borrowed), Date Borrowed, and Date Returned.
- The museum keeps track of ARTIST's information: Name, Date Born, Main Style, and description. The name assumed to be unique.
- Different EXHIBITIONS occur, each having a Name, Start Date, and End Date. EXHIBITIONS are related to all the art objects that were on display during the exhibition.
- The EXHIBITIONS can be CHARITABLE or COMMERCIAL. The CHARITABLE exhibition has Sponsors. For each COMMERCIAL exhibition has specific Fees for each meter and Space of reservation. EXHIBITIONS can be mix of both types.

2a. Insert the following keys 4, 8, 39, 16, 10, 54, 65, 90, 76, and 19 (strictly in this order) into an initially empty B-Tree of order 3 ($m=3$). Illustrates the steps of insertion of each element. [4 M]

2b. Consider B+ tree in which the search key is 12 bytes long, block size is 1024 bytes, record pointer is 10 bytes long and block pointer is 8 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is. Show all intermediate steps. [3 M]

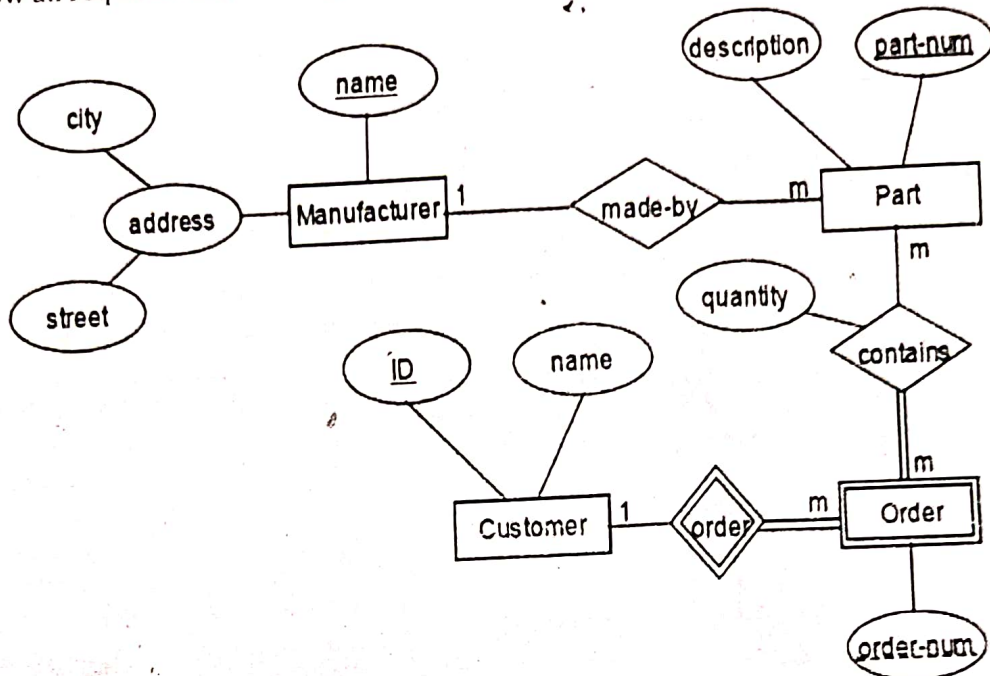
3. An Enterprise wishes to maintain a database to automate its operations. Enterprise is divided into certain departments and each department consist of employees. The following two tables describes the automation schemas.

Dept (deptno, dname, loc)

Emp (empno, ename, street, city, job, hiredate, sal, deptno)

- a) Update the employee salary by 15%, whose experience is greater than 30 years. [1 M]
- b) Determine the department which does not contain any employees. [1 M]
- c) Display the details of those employee who draw salary greater than the average salary. [2 M]
- d) Find the department that has the highest number of employees. [2 M]

4. Find the minimum number of tables required to represent the given ER diagram in relational model. Show all required tables along with their corresponding attributes and keys. [4 M]



5. Consider a relation schema $R(A, B, C, D, E, F, G, H)$ with the following set of functional dependencies:

$A \rightarrow BCD$
 $CD \rightarrow G$
 $B \rightarrow F$
 $BC \rightarrow E$
 $DG \rightarrow H$

Prove that $A \rightarrow FG$, and $AC \rightarrow E$ can be additionally derived from the above set of functional dependencies using Inference rules of functional dependencies. [4 M]

$$12 \times 8 = 96$$

$$12 \times 8 + (12-1)K \leq 1024$$

$$96 + 11K \leq 1024$$

$$11K \leq 1024 - 96$$

$$\frac{928}{11}$$

$$\begin{array}{r} 1024 \\ - 96 \\ \hline 928 \end{array}$$

$$\begin{array}{r} 8 \\ 928 \\ \hline 11 \end{array}$$